

Question\_6:

Code:

```
1  /*
2  Question_6: Find the third largest element in the array.
3  */
4
5  #include <iostream>
6  #include <climits>
7  using namespace std;
8
9  void thirdLargest(int arr[], int arr_size) {
10     int a = INT_MIN, b = INT_MIN, c = INT_MIN;
11
12     for (int i = 0; i < arr_size; i++) {
13         if (arr[i] > a) {
14             c = b;
15             b = a;
16             a = arr[i];
17         } else if (arr[i] > b && arr[i] != a) {
18             c = b;
19             b = arr[i];
20         } else if (arr[i] > c && arr[i] != b && arr[i] != a) {
21             c = arr[i];
22         }
23     }
24     cout << "The third Largest element is " << c;
25 }
26 int main() {
27     int n;
28     cin >> n;
29
30     int arr[1000];
31     for (int i = 0; i < n; i++)
32         cin >> arr[i];
33
34     thirdLargest(arr, n);
35     return 0;
36 }
```

**Output:**

**Test Case\_1:**

```
6
1 14 2 16 10 20
The third Largest element is 14
PS C:\Users\Prabin Yadav\Document
```

**Test Case\_2:**

```
7
19 -10 20 14 2 16 10
The third Largest element is 16
PS C:\Users\Prabin Yadav\Documents\
```